# Generalized Probabilistic Theories

Operational foundations, convex geometry and the reconstruction of quantum theory

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***Guiding questions.***

*(i) What is a probabilistic theory, if we assume no Hilbert space, density matrix or Born rule?*

*(ii) Within this framework, what distinguishes classical theory, quantum theory and boxworld?*

*(iii) Which operational principles single out complex quantum theory?*

*Chapters 0–2 answer (i), Chapter 3 answers (ii), and Chapter 4 answers (iii).*

***Levels of assumption.*** *Throughout, we keep three kinds of assumption apart:*

- framework axioms*: mixing, closure, finite dimension;*
- optional hypotheses*: no-restriction, tomographic locality;*
- reconstruction principles*: subspace axiom, continuous reversibility.*

*Every theorem lists the assumptions it uses.*

# Chapter 0 Operational Foundations

## 0.1 Procedures and operational equivalence

**Definition 0.1.1 (Operational scenario):** A *preparation* $\mathbf{P}$, an optional *transformation* $\mathbf{T}$ and a *measurement* $\mathbf{M}$ with finitely many outcomes $i$ define outcome probabilities $p(i|\mathbf{P},\mathbf{T},\mathbf{M})$. A transformation followed by a measurement is again a measurement, so it suffices to consider $p(i|\mathbf{P},\mathbf{M})$.

**Definition 0.1.2 (Operational equivalence, states):** $\mathbf{P} \simeq \mathbf{P}'$ iff $p(i|\mathbf{P},\mathbf{M}) = p(i|\mathbf{P}',\mathbf{M})$ for all measurements $\mathbf{M}$ and outcomes $i$. A *state* is an equivalence class of preparations. Dually, two measurement outcomes $(\mathbf{M},i)$ and $(\mathbf{M}',i')$ are equivalent iff they have equal probabilities on all preparations; an *effect* is an equivalence class of outcomes.

**Remark 0.1.3:** A state carries all, and only, the measurement statistics. Different physical procedures can prepare the same state; e.g. in quantum theory, an equal mixture of $|\uparrow\rangle, |\downarrow\rangle$ and half of a Bell pair both give $\mathbb{I}/2$.

## 0.2 Classical randomisation

**Definition 0.2.1 (Framework axioms):** (F1) *Mixing:* for states $\omega_1, \omega_2$ and $t \in [0,1]$, the procedure "prepare $\omega_1$ with probability $t$, else $\omega_2$, and forget the choice" is a preparation. Its outcome probabilities are $t\,p(i|\omega_1,\mathbf{M}) + (1-t)\,p(i|\omega_2,\mathbf{M})$.

(F2) *Closure:* limits of preparable states are preparable.

(F3) *Finite dimension:* the states span a finite-dimensional real vector space.

**Proposition 0.2.2 (States form a compact convex set):** Under (F1)–(F3), the set of states embeds affinely as a compact convex subset $\Omega$ of a finite-dimensional real vector space. Every measurement outcome defines an affine function $\Omega \to [0,1]$.

*Proof.* Map each state to its list of outcome probabilities, $\omega \mapsto (p(i|\omega,\mathbf{M}))_{i,\mathbf{M}} \in [0,1]^{\mathcal{I}}$. This map is injective by operational equivalence, and by (F1) it sends mixtures to convex combinations. So $\Omega$ is a convex subset of $[0,1]^{\mathcal{I}}$, hence bounded. Restrict to the span, which is finite-dimensional by (F3). Then $\Omega$ is closed by (F2), hence compact. Each outcome is a coordinate function, which is affine by (F1). □

**Remark 0.2.3:**

- Boundedness is automatic: probabilities lie in $[0, 1]$.
- Closure and finite dimension are idealisations: (F2) is motivated by finite experimental precision; (F3) is a genuine restriction (Plávala, Def. 2.1).
- Mixing forgets the classical label. Keeping it is modelled by a classical register (Chapter 2).

**Summary of Chapter 0.** Operational states are equivalence classes of preparations, and classical randomisation makes the state space convex.

# Chapter 1 State Spaces and Effects

## 1.1 Ordered vector space formulation

**Definition 1.1.1 (State space):** A *state space* is a triple $(V, V_+, u)$ where:

- $V$ is a finite-dimensional real vector space;
- $V_+ \subset V$ is a closed, convex, pointed ($V_+ \cap -V_+ = \{0\}$) and generating ($V_+ - V_+ = V$) cone;
- $u \in V^*$ is strictly positive on $V_+ \setminus \{0\}$.

The *normalised states* are $\Omega = \{\omega \in V_+ : u(\omega) = 1\}$; the elements of $V_+$ are unnormalised (e.g. subnormalised) states.

**Lemma 1.1.2:** (i) $\Omega$ is compact and convex, and $\dim \operatorname{aff}(\Omega) = \dim V - 1$.

(ii) $\Omega$ is a *base* of $V_+$: every $v \in V_+ \setminus \{0\}$ equals $\lambda\omega$ for unique $\lambda > 0$ and $\omega \in \Omega$.

(iii) Conversely, every compact convex $\Omega_0$ arises in this way, with $V = \operatorname{aff}(\Omega_0) \oplus \mathbb{R}$ and $\omega \mapsto (\omega, 1)$.

*Proof.* (ii) Take $\lambda = u(v) > 0$ and $\omega = v/u(v)$.

(i) Since $u$ is strictly positive and continuous on the closed cone $V_+$, it attains a minimum $c > 0$ on $V_+ \cap \{\|v\| = 1\}$. Hence $\|v\| \leqslant u(v)/c$, so $\Omega$ is bounded; it is closed and convex. The dimension statement follows from (ii), since $V_+$ is generating.

(iii) Take $V_+ = \operatorname{cone}(\Omega_0 \times \{1\})$ and $u(x, t) = t$ (Plávala, Prop. 3.20). □

**Definition 1.1.3 (Effects and measurements):**

- The *dual cone* is $V_+^* = \{e \in V^* : e(v) \geqslant 0 \ \forall v \in V_+\}$.
- The *effects* are $E = [0, u] = \{e \in V^* : 0 \leqslant e \leqslant u\} = \{e \in V^* : 0 \leqslant e(\omega) \leqslant 1 \ \forall \omega \in \Omega\}$.
- A *measurement* is a finite family $(e_i)$ in $E$ with $\sum_i e_i = u$.
- The *complement* of $e$ is $e^\perp = u - e \in E$.

**Remark 1.1.4 (Physical meaning):** By Proposition 0.2.2, outcome probabilities are affine in the state, so they extend to linear functionals on $V$. The constraint $\sum_i e_i = u$ is normalisation of outcome probabilities, which must hold on all of $\Omega$ and hence, since $\Omega$ spans $V$, on all of $V$.

## 1.2 Duality

**Theorem 1.2.1: Bipolar duality**

For a closed convex cone, $(V_+^*)^* = V_+$ under the identification $V^{**} = V$. Consequently,

$$\Omega = \{\omega \in V : e(\omega) \geqslant 0 \ \forall e \in E,\ u(\omega) = 1\},$$

i.e. the states are exactly the normalised functionals that are positive on effects, and conversely.

*Proof.* $V_+ \subseteq (V_+^*)^*$ is immediate. Suppose $w \in (V_+^*)^* \setminus V_+$. By strict separation of the point $w$ from the closed convex cone $V_+$ (Appendix A), there is $e \in V^*$ with $e(V_+) \geqslant 0 > e(w)$. Then $e \in V_+^*$, contradicting $w \in (V_+^*)^*$. The description of $\Omega$ follows because $E$ generates $V_+^*$ as a cone ($e \in V_+^*$ implies $e/\max_\Omega e \in E$). □

**Definition 1.2.1 (Self-duality):** $(V, V_+)$ is *strongly self-dual* if there is an inner product on $V$ with $V_+^* = V_+$ under the induced identification $V^* \cong V$.

## 1.3 Pure states, faces and distinguishability

**Definition 1.3.1:**

- $\omega \in \Omega$ is *pure* if it is an extreme point of $\Omega$, and *mixed* otherwise.
- A *face* $F \subseteq \Omega$ is a convex subset with: $t\omega_1 + (1-t)\omega_2 \in F$, $t \in (0,1) \Rightarrow \omega_1, \omega_2 \in F$.
- $\Omega$ is *strictly convex* if its only faces are $\emptyset$, points and $\Omega$ itself; equivalently, its topological boundary (relative to $\operatorname{aff}\Omega$) contains no segment.

**Theorem 1.3.1: Minkowski–Carathéodory**

$\Omega = \operatorname{conv}(\operatorname{ext}\Omega)$. Every $\omega \in \Omega$ is a convex combination of at most $\dim V$ pure states.

**Remark 1.3.2:** The decomposition into pure states is unique for all states iff $\Omega$ is a simplex. Non-uniqueness is a generic feature, not a special trait of quantum theory.

**Definition 1.3.3 (Distinguishability, capacity, dimension):**

- $\omega_1, \dots, \omega_n$ are *jointly perfectly distinguishable* if there is a measurement $(e_i)_{i=1}^n$ with $e_i(\omega_j) = \delta_{ij}$.
- The *capacity* $N$ is the largest such $n$.
- The *dimension* is $K = \dim V$.

## 1.4 Norms and state discrimination

**Definition 1.4.1 (Order-unit and base norms):**

- *Order-unit norm* on $V^*$: $\|e\|_u = \inf\{\lambda \geqslant 0 : -\lambda u \leqslant e \leqslant \lambda u\} = \max_{\omega\in\Omega} |e(\omega)|$.
- *Base norm* on $V$: $\|v\|_\Omega = \inf\{\alpha + \beta : v = \alpha\omega_1 - \beta\omega_2,\ \alpha, \beta \geqslant 0,\ \omega_i \in \Omega\}$.

**Lemma 1.4.2:** The base norm is the dual of the order-unit norm: $\|v\|_\Omega = \sup\{|e(v)| : \|e\|_u \leqslant 1\} = \sup_{e\in E} |(2e - u)(v)|$.

*Proof (second equality).* $\|g\|_u \leqslant 1$ iff $-u \leqslant g \leqslant u$, iff $e := \frac{1}{2}(u+g) \in E$. The first equality (base norm = dual norm) is standard; see Plávala, Prop. 3.42. □

**Theorem 1.4.1: Optimal discrimination of two states**

States $\omega_0, \omega_1$ are given with prior probabilities $\lambda, 1-\lambda$. Under the no-restriction hypothesis,

the optimal success probability of a two-outcome measurement is

$$p_{\text{succ}} = \tfrac{1}{2}\big(1 + \|\lambda\omega_0 - (1-\lambda)\omega_1\|_\Omega\big).$$

*Proof.* With the effect $e$ for the guess "$\omega_0$",

$$p(e) = \lambda e(\omega_0) + (1-\lambda)(1 - e(\omega_1)) = (1-\lambda) + e(\psi), \qquad \psi := \lambda\omega_0 - (1-\lambda)\omega_1.$$

Write $e = \frac{1}{2}(u + g)$ with $\|g\|_u \leqslant 1$. Since $u(\psi) = 2\lambda - 1$,

$$p(e) = \tfrac{1}{2} + \tfrac{1}{2}g(\psi).$$

Maximise over $g$ using the lemma, and note that $g \mapsto -g$ preserves the unit ball. □

**Remark 1.4.3:** The base norm is therefore *operational*. In quantum theory it is the trace norm, and the theorem is Helstrom's bound. Optimisation over $E$ uses the no-restriction hypothesis; in a restricted theory, $E$ is replaced by the allowed effects.

## 1.5 No-restriction hypothesis and restricted theories

**Definition 1.5.1 (No-restriction hypothesis (NRH)):** Every $e \in E = [0, u]$ is a physically implementable effect.

**Definition 1.5.2 (Restricted theory):** A set $E_{\text{phys}} \subseteq E$ of allowed effects, assumed convex, containing 0 and $u$, closed under $e \mapsto u - e$, and separating the states. Its dual state space

$$S(E_{\text{phys}}) = \{v : e(v) \geqslant 0 \ \forall e \in E_{\text{phys}},\ u(v) = 1\}$$

satisfies $\Omega \subseteq S(E_{\text{phys}})$.

**Remark 1.5.3:**

- NRH is an *extra hypothesis*, not part of the framework. Stabilizer quantum mechanics and Spekkens' toy theory are restricted.
- Consistency requires restrictions to be compatible with composition and transformations (Janotta–Lal).
- The distinction matters for generalized contextuality: noncontextuality concerns the *allowed* effects.

## 1.6 Equivalent state spaces

**Definition 1.6.1:** $(V, V_+, u) \simeq (W, W_+, u')$ if there is a linear isomorphism $L$ with $L(V_+) = W_+$ and $u' \circ L = u$. Dynamical data transform by $T \mapsto LTL^{-1}$, and effects by $e \mapsto e \circ L^{-1}$. All probabilities are invariant.

**Remark 1.6.2 (Fiducial effects):** Choose a basis $e_1, \ldots, e_K$ of $V^*$ consisting of effects; e.g. shrink any basis into $E$ using the order unit. Then $\omega \mapsto (e_k(\omega))_k$ represents states as lists of probabilities (Hardy, Barrett). This representation is equivalent to the abstract one.

# Chapter 2 Transformations and Composite Systems

## 2.1 Transformations

**Proposition 2.1.1 (Transformations are linear):** Assume a transformation commutes with mixing: preparing $\omega_i$ with probability $t_i$ and then applying $T$ is the same procedure as applying $T$ to the mixture. Then $T$ extends uniquely to a linear map $V_A \to V_B$ with $T(V_{A,+}) \subseteq V_{B,+}$. If $T$ is deterministic, then also $u_B \circ T = u_A$.

*Proof.* $T$ is affine on $\Omega_A$ by hypothesis. Extend it by homogeneity to $V_{A,+}$, and by linearity to $V_A$, using that $V_{A,+}$ is generating. Positivity holds because states map to states; deterministic maps preserve normalisation. □

**Definition 2.1.2 (Transformations, reversibility, dynamical state space):**

- A *(deterministic) transformation* is a positive linear map $T$ with $u_B \circ T = u_A$.
- It is *reversible* if $T^{-1}$ exists and is a transformation; equivalently, $T(\Omega) = \Omega$, i.e. $T$ is a linear symmetry of $\Omega$.
- A *dynamical state space* $(V, V_+, u, \mathcal{T})$ specifies a compact group $\mathcal{T}$ of *allowed* reversible transformations.

**Remark 2.1.3 (Positive $\neq$ allowed):** Positivity on a single system is necessary but not sufficient. A transformation must remain valid when applied to part of a composite system in the theory:

$$(\mathrm{id}_C \otimes T)\big(\Omega_{CA}\big) \subseteq \Omega_{CB} \quad \text{for all systems } C \text{ of the theory.}$$

This is the GPT analogue of complete positivity, and it depends on the chosen composites. The set of allowed channels is therefore part of the *data* of a theory. Example: quantum transposition is positive but not allowed.

**Definition 2.1.4 (Measurements and instruments as channels):**

- A measurement $(e_i)_{i=1}^n$ is the channel $\omega \mapsto (e_i(\omega))_i$ into the classical simplex $\Delta_n$.
- An *instrument* is a family $(T_i)$ of positive maps with $\sum_i u_B \circ T_i = u_A$.

## 2.2 Composite systems

**Definition 2.2.1 (Composition axioms (Plávala, Def. 5.1)):** A composite $\Omega_{AB}$ of $\Omega_A$ and $\Omega_B$ satisfies:

(C1) $\Omega_{AB}$ is a state space;

(C2) independent preparations: for all $\omega_A, \omega_B$ there is a product state $\omega_A \otimes \omega_B \in \Omega_{AB}$, biaffine in $(\omega_A, \omega_B)$;

(C3) independent measurements: product effects $e_A \otimes e_B$ are effects on $\Omega_{AB}$, bilinear in $(e_A, e_B)$, with $(e_A \otimes e_B)(\omega_A \otimes \omega_B) = e_A(\omega_A) e_B(\omega_B)$;

(C4) $u_{AB} = u_A \otimes u_B$;

(TL) *tomographic locality:* states of $\Omega_{AB}$ are determined by their values on product effects.

**Lemma 2.2.2 (Local tomography fixes the vector space):** Under (C2), (C3) and (TL), $V_{AB} \cong V_A \otimes V_B$. Equivalently, $K_{AB} = K_A K_B$.

*Proof.* By (C3), every $\omega \in \Omega_{AB}$ defines the bilinear functional $(e_A, e_B) \mapsto (e_A \otimes e_B)(\omega)$, i.e. an element $\hat{\omega} \in (V_A^* \otimes V_B^*)^* = V_A \otimes V_B$. By (TL), $\omega \mapsto \hat{\omega}$ is injective. On product states, $\widehat{\omega_A \otimes \omega_B} = \omega_A \otimes \omega_B$, and these span $V_A \otimes V_B$; so the induced linear map $V_{AB} \to V_A \otimes V_B$ is a bijection.

For the dimension count: without (TL), the same map is still *surjective*, because it is onto on product states, so $K_{AB} \geqslant K_A K_B$. Equality holds iff the map is injective, i.e. iff (TL) holds. □

**Definition 2.2.3 (Minimal and maximal tensor products):**

$$\Omega_A \otimes_{\min} \Omega_B = \operatorname{conv}\{\omega_A \otimes \omega_B : \omega_A \in \Omega_A,\ \omega_B \in \Omega_B\},$$
$$\Omega_A \otimes_{\max} \Omega_B = \{\omega \in V_A \otimes V_B : (e_A \otimes e_B)(\omega) \geqslant 0\ \forall e_A \in E_A,\ e_B \in E_B,\ (u_A \otimes u_B)(\omega) = 1\}.$$

States in $\otimes_{\min}$ are *separable*; states outside it are *entangled*.

**Theorem 2.2.1: Range of composites**

Under (C1)–(C4) and (TL):

$$\Omega_A \otimes_{\min} \Omega_B \subseteq \Omega_{AB} \subseteq \Omega_A \otimes_{\max} \Omega_B.$$

Dually, $E_A \otimes_{\min} E_B \subseteq E_{AB} \subseteq E_A \otimes_{\max} E_B$. Conversely, every compact convex set in between satisfies (C1)–(C4) and (TL). So, in general, the composite is *not* determined by $\Omega_A$ and $\Omega_B$: it is additional data of the theory.

*Proof.* The left inclusion follows from (C2) and convexity. For the right inclusion: by (C3), product effects are effects, so they are non-negative on $\Omega_{AB}$; by (C4), states are normalised; by the lemma, $\Omega_{AB} \subset V_A \otimes V_B$. The dual statement follows from the definition of $E$ as $[0, u]$. □

**Remark 2.2.4:**

- Under the NRH, the maximal composite allows only separable effects, and the minimal composite allows all effects that are positive on product states.
- Both $\otimes_{\min}$ and $\otimes_{\max}$ are associative (Plávala, Props. 5.11, 5.13). Multipartite theories require associativity and the coherence conditions of a symmetric monoidal category.

## 2.3 Consequences of the composition axioms

**Theorem 2.3.1: No-signalling**

Under (C3) and (C4), local measurements $\{e_x^{(a)}\}_a$ and $\{f_y^{(b)}\}_b$ on any $\omega \in \Omega_{AB}$ give non-signalling statistics $p(a, b|x, y) = (e_x^{(a)} \otimes f_y^{(b)})(\omega)$.

*Proof.* $\sum_b p(a, b|x, y) = (e_x^{(a)} \otimes u_B)(\omega)$ is independent of $y$, and symmetrically for $x$. □

**Definition 2.3.1 (Conditional and reduced states):** For $\omega \in \Omega_{AB}$ and $f \in E_B$, the *conditional (unnormalised) state* is $(\mathrm{id} \otimes f)(\omega) \in V_A$. The *reduced state* is $\omega_A = (\mathrm{id} \otimes u_B)(\omega)$; this generalises the partial trace.

**Lemma 2.3.2:** If $\omega \in \Omega_A \otimes_{\max} \Omega_B$, then $(\mathrm{id} \otimes f)(\omega) \in V_{A,+}$, and it equals $\big((u_A \otimes f)(\omega)\big)\,\omega_f$ for some $\omega_f \in \Omega_A$ whenever $(u_A \otimes f)(\omega) > 0$.

*Proof.* For every $e \in E_A$, $e\big((\mathrm{id} \otimes f)(\omega)\big) = (e \otimes f)(\omega) \geqslant 0$, so $(\mathrm{id} \otimes f)(\omega) \in (V_{A,+}^*)^* = V_{A,+}$ by Theorem 1.2.1. Normalise using $u_A$. □

**Theorem 2.3.2: Pure marginals force product states**

Assume (C1)–(C4) and (TL). If $\omega \in \Omega_{AB}$ has a *pure* reduced state $\omega_A$, then $\omega = \omega_A \otimes \omega_B$.

*Proof.* Fix $f \in E_B$ and put $p = (u_A \otimes f)(\omega)$. By the lemma,

$$\omega_A = (\mathrm{id} \otimes f)(\omega) + (\mathrm{id} \otimes (u - f))(\omega) = p\,\omega_f + (1-p)\,\omega_{f^\perp}, \qquad \omega_f, \omega_{f^\perp} \in \Omega_A.$$

By purity, $\omega_f = \omega_A$ whenever $p > 0$. Hence $(\mathrm{id} \otimes f)(\omega) = p\,\omega_A$ for every $f$ (trivially also when $p = 0$). Therefore, for all $e \in E_A$ and $f \in E_B$,

$$(e \otimes f)(\omega) = e(\omega_A)\, f(\omega_B) = (e \otimes f)(\omega_A \otimes \omega_B).$$

By (TL), $\omega = \omega_A \otimes \omega_B$. □

**Remark 2.3.3:** This is a GPT form of the monogamy of entanglement: a system in a pure state cannot be correlated with anything. Without (TL) the statement can fail.

**Lemma 2.3.4 (Capacity is supermultiplicative):** $N_{AB} \geqslant N_A N_B$.

*Proof.* If $(e_i)$ distinguishes $(\omega_k)$ and $(f_j)$ distinguishes $(\sigma_l)$, then $(e_i \otimes f_j)(\omega_k \otimes \sigma_l) = \delta_{ik}\delta_{jl}$. □

# Chapter 3 The GPT Landscape: Three Fundamental Theories

## 3.1 Classical probability theory

**Definition 3.1.1:** $V = \mathbb{R}^N$, $V_+ = \mathbb{R}^N_+$, $u = (1, \ldots, 1)$, so that $\Omega = \Delta_N$ is the simplex.

**Proposition 3.1.2:** (i) The pure states are the $N$ vertices; every state has a *unique* pure decomposition; $N = K$.

(ii) $E = [0,1]^N$, and $V_+$ is strongly self-dual (dot product).

(iii) The reversible transformations are exactly the permutations of the vertices.

(iv) $\Delta_M \otimes_{\min} \Delta_N = \Delta_M \otimes_{\max} \Delta_N = \Delta_{MN}$: the composite is unique.

*Proof.* (i)–(ii) are direct.

(iii) A linear symmetry of $\Omega$ maps extreme points to extreme points, and is determined by its values on the vertex basis.

(iv) $\omega \in \otimes_{\max}$ is non-negative on all $b_i \otimes b_j$ (the products of vertex indicators) and normalised. So it is a probability vector on $[M] \times [N]$, hence a mixture of the products of vertices. □

**Theorem 3.1.1: Entangleability (Aubrun–Lami–Palazuelos–Plávala)**

$\Omega_A \otimes_{\min} \Omega_B = \Omega_A \otimes_{\max} \Omega_B$ iff $\Omega_A$ or $\Omega_B$ is a simplex. So every pair of non-classical systems admits entangled states in some composite.

## 3.2 Quantum theory

**Definition 3.2.1:** $V = \mathrm{Herm}(\mathbb{C}^N)$ (a *real* vector space, $K = N^2$), $V_+ = \{\rho \geqslant 0\}$, $u = \mathrm{tr}$.

**Proposition 3.2.2:** (i) With the Hilbert–Schmidt pairing, $E = \{0 \leqslant M \leqslant \mathbb{I}\}$, measurements are POVMs, and $V_+^* = V_+$ (strong self-duality).

(ii) The pure states are the rank-one projectors; $N$ equals the Hilbert-space dimension.

(iii) The linear symmetries of $\Omega$ are $\rho \mapsto U\rho U^\dagger$ with $U$ unitary or antiunitary (Wigner). Only unitaries are allowed once composites are included, since antiunitaries are not completely positive.

(iv) The quantum composite $\mathrm{Herm}(\mathbb{C}^{MN})$ satisfies (TL), and lies strictly between $\otimes_{\min}$ (separable states) and $\otimes_{\max}$ (which contains entanglement witnesses).

(v) For $N = 2$, $\Omega$ is affinely the Bloch ball:

$$\rho = \tfrac{1}{2}\begin{pmatrix} 1+r_3 & r_1 - ir_2 \\ r_1 + ir_2 & 1 - r_3 \end{pmatrix}, \qquad |r| \leqslant 1.$$

*Proof of (i).* $\mathrm{tr}(M\rho) \in [0,1]$ for all density matrices iff $\langle\psi|M|\psi\rangle \in [0,1]$ for all unit vectors, i.e. $0 \leqslant M \leqslant \mathbb{I}$. Self-duality: $\mathrm{tr}(AB) \geqslant 0$ for all $B \geqslant 0$ iff $A \geqslant 0$. □

**Remark 3.2.3:** For $N \geqslant 3$, $\Omega$ is not strictly convex (e.g. $\mathrm{diag}(\frac{1}{2}, \frac{1}{2}, 0)$ is a mixed boundary point), so it is not a ball. On pure states, the dynamics $|\psi\rangle\langle\psi| \mapsto U|\psi\rangle\langle\psi|U^\dagger$ can be "square-rooted" to $|\psi\rangle \mapsto U|\psi\rangle$: the superposition principle is a consequence of the shape of $\Omega$.

## 3.3 Boxworld

**Definition 3.3.1 (Gbit):** $V = \mathbb{R}^3$, $\Omega = \{(x, y, 1) : |x|, |y| \leqslant 1\}$ (a square), $u = (0, 0, 1)$. The effects are $\mathrm{conv}\{0, u, e^{(x)}, \bar{e}^{(x)}, e^{(y)}, \bar{e}^{(y)}\}$, where

$$e^{(x)} = \tfrac{1}{2}(1,0,1), \quad \bar{e}^{(x)} = \tfrac{1}{2}(-1,0,1), \quad e^{(y)} = \tfrac{1}{2}(0,1,1), \quad \bar{e}^{(y)} = \tfrac{1}{2}(0,-1,1).$$

**Proposition 3.3.2:** (i) Any two distinct pure states are perfectly distinguishable.

(ii) No three states are jointly perfectly distinguishable, so $N = 2$.

(iii) The gbit is not strongly self-dual.

*Proof.* (i) Adjacent vertices differ in one coordinate and are distinguished by $(e^{(x)}, \bar{e}^{(x)})$ or $(e^{(y)}, \bar{e}^{(y)})$. Opposite vertices differ in both coordinates and are distinguished by either measurement.

(ii) Any three vertices contain a diagonal pair $\omega_a, \omega_c$, and the third vertex $\omega_b$ lies on the other diagonal $\{\omega_b, \omega_d\}$. The centre is $m = \frac{1}{2}(\omega_a + \omega_c) = \frac{1}{2}(\omega_b + \omega_d)$. If $(e_1, e_2, e_3)$ distinguished $\omega_a, \omega_b, \omega_c$, then

$$e_b(m) = \tfrac{1}{2}\big(e_b(\omega_a) + e_b(\omega_c)\big) = 0, \qquad \text{but also} \qquad e_b(m) = \tfrac{1}{2}\big(1 + e_b(\omega_d)\big) \geqslant \tfrac{1}{2}.$$

For general states: if $e_i(\omega_j) = \delta_{ij}$, then choose one pure state in a decomposition of each $\omega_j$. Since $0 \leqslant e_i \leqslant u$, every pure component of $\omega_j$ satisfies $e_i = \delta_{ij}$. This gives three distinct, jointly distinguishable vertices, which is impossible by the above.

(iii) See Janotta–Gogolin–Barrett–Brunner (2011); no inner product makes the two cones coincide. □

**Theorem 3.3.1: Boxworld and the no-signalling polytope**

gbit $\otimes_{\max}$ gbit is affinely isomorphic to the polytope $\mathcal{NS}_{2,2,2}$ of non-signalling behaviours. Its 24 extreme points are the 16 products of pure states and 8 PR boxes. In particular, CHSH reaches 4, beyond the quantum bound $2\sqrt{2}$ and the classical bound 2.

*Proof sketch.* The extreme effects of a gbit are the two binary measurements. So positivity on product effects is exactly positivity of the 16 probabilities $p(a, b|x, y)$, and normalisation together with (C4) gives the non-signalling constraints (Theorem 2.3.1). For the classical bound: for deterministic $a_x, b_y \in \{\pm 1\}$, $a_0(b_0 + b_1) + a_1(b_0 - b_1) = \pm 2$. □

**Remark 3.3.3:** In boxworld the reversible dynamics is *trivial*: local operations and permutations of subsystems only (Gross et al.). Entanglement can never be created reversibly.

## 3.4 Structural results

**Theorem 3.4.1: No-broadcasting (Barnum–Barrett–Leifer–Wilce)**

A universal broadcasting map $B : V \to V \otimes V$ (positive, normalised, into an allowed composite) with $(\mathrm{id} \otimes u) \circ B = (u \otimes \mathrm{id}) \circ B = \mathrm{id}$ exists iff $\Omega$ is a simplex.

*Proof of "if".* On a simplex, define $B(s_i) = s_i \otimes s_i$ on the vertices and extend linearly. $B$ is positive, maps into $\otimes_{\min}$, and both marginals of $s_i \otimes s_i$ are $s_i$. The converse ("only if") is the theorem of Barnum–Barrett–Leifer–Wilce; it is not reproduced here. □

**Remark 3.4.1:**

- *Self-duality:* strong self-duality follows from *bit symmetry* (any pair of pure distinguishable states maps reversibly to any other such pair; Müller–Ududec). It holds in classical and quantum theory but fails for the gbit.
- *Generic features:* entanglement (Theorem 3.1.1), no-broadcasting and no-signalling are *generic* non-classical features of GPTs. They do not single out quantum theory.

# Chapter 4 Reconstruction of Quantum Theory

## 4.1 Assumptions and main theorem

**Background assumptions.**

- finite dimension (F3) and the NRH;
- tomographic locality (TL) for all composites;
- for every $N \in \mathbb{N}$, a dynamical state space $(V_N, \Omega_N, \mathcal{T}_N)$ of capacity $N$;
- composites of these systems are again systems of the theory.

**Reconstruction principles.**

- **Subspace axiom (SA).** If $e_1, \ldots, e_N$ perfectly distinguishes $N$ states of $\Omega_N$, then $\{\omega \in \Omega_N : e_N(\omega) = 0\} \simeq \Omega_{N-1}$, and the transformations in $\mathcal{T}_N$ preserving this face act as $\mathcal{T}_{N-1}$.
- **Continuous reversibility (CR).** For every system of the theory (in particular every $\Omega_N$ and every composite), the allowed reversible group is a connected compact group acting transitively on the pure states.

**Theorem 4.1.1: Masanes–Müller**

Under the background assumptions, (SA) and (CR), for every $N$:

$$V_N \simeq \mathrm{Herm}(\mathbb{C}^N), \qquad \Omega_N \simeq \{\rho \geqslant 0,\ \mathrm{tr}\,\rho = 1\}, \qquad \mathcal{T}_N \simeq \{\rho \mapsto U\rho U^\dagger : U \in U(N)\}.$$

**Remark 4.1.1:** None of complex numbers, operators, or an algebraic product is assumed. The proof below is complete for capacity two up to dimension (Section 4.2), and is a *proof sketch* beyond that; see Masanes–Müller (2011) for details.

## 4.2 Why the qubit is a ball

**Lemma 4.2.1:** A state space with more than one state has capacity at least 2. Hence $\Omega_1$ is a single point.

*Proof.* Let $\Omega$ contain two states, so $d = \dim \operatorname{aff} \Omega \geqslant 1$. A compact convex set has an exposed point $\omega_1$ (Straszewicz): there is an affine functional $\ell$ with $\ell(\omega_1) = 1 > \ell(\omega)$ for all $\omega \neq \omega_1$. Rescale and shift $\ell$ so that $\min_\Omega \ell = 0$, attained at some $\omega_2$, and $\max_\Omega \ell = 1$. The affine function $\ell$ extends to $e_1 \in V^*$ with $0 \leqslant e_1 \leqslant u$, which is an effect by the NRH. Then $(e_1, u - e_1)$ distinguishes $\omega_1$ and $\omega_2$. □

**Lemma 4.2.2:** Under (SA), $\Omega_2$ is strictly convex.

*Proof.* If $\Omega_2$ is not strictly convex, it has a proper face with more than one point. The smallest exposed face containing that face is a proper exposed face $F$ with more than one point. So there is a non-constant affine $\ell$ with $\max_\Omega \ell = 1$ and $F = \{\ell = 1\} \cap \Omega_2$; shift and rescale so that $\min_\Omega \ell = 0$. As in the previous proof, $(e_1, e_2) = (\ell, u - \ell)$ perfectly distinguishes any $\omega_1 \in F$ from any $\omega_2$ with $\ell(\omega_2) = 0$. By (SA), $\{e_2 = 0\} = F$ is equivalent to $\Omega_1$, a single point. Contradiction. □

**Theorem 4.2.1: Bits are balls**

Under (SA) and (CR), $\Omega_2$ is affinely a Euclidean unit ball $B^d$, and $\mathcal{T}_2$ acts by orthogonal transformations.

*Proof.* (i) *Unique invariant state.* For pure $\omega$, set $\mu = \int_{\mathcal{T}_2} T\omega \, dT$ (Haar). By transitivity, $\mu$ does not depend on $\omega$. If $\tau = \sum_i p_i \omega_i$ is invariant, then $\tau = \int T\tau \, dT = \sum_i p_i \mu = \mu$.

(ii) *Orthogonality.* On $W = \operatorname{aff}(\Omega_2) - \mu$, average any inner product over $\mathcal{T}_2$; the $T$ become orthogonal.

(iii) *Pure states on a sphere.* By transitivity, all pure states have the same norm, normalised to 1. So $\operatorname{ext} \Omega_2 \subseteq S^{d-1}$ and $\Omega_2 \subseteq B^d$.

(iv) *No gaps.* Suppose $\Omega_2 \neq B^d$. Pick $x \in S^{d-1} \setminus \Omega_2$; the segment from $\mu$ to $x$ leaves $\Omega_2$ at a boundary point $y$ with $|y| < 1$. By strict convexity, every boundary point is extreme (otherwise the boundary would contain a segment), so $y \in \operatorname{ext} \Omega_2 \subseteq S^{d-1}$. Contradiction. □

## 4.3 Why the ball is three-dimensional

**Lemma 4.3.1:** $d \geqslant 2$.

*Proof.* If $d = 1$, then $\Omega_2$ is a segment, i.e. a classical bit. By Proposition 3.1.2(iv), the composite of two such bits is unique and equals $\Delta_4$. Since composites are systems of the theory, (CR) applies to it. But its reversible transformations are the permutations of 4 vertices, a finite group, and a connected group acting on a finite set acts trivially. This contradicts transitivity. □

*Proof sketch of $d = 3$.* (i) *Capacity and dimension.* The axioms imply $N_{AB} = N_A N_B$ (Masanes–Müller), and (TL) gives $K_{AB} = K_A K_B$. Hence $K = N^r$ for some $r \in \mathbb{N}$ (Hardy), so $d = K_2 - 1 = 2^r - 1 \in \{1, 3, 7, 15, \dots\}$.

(ii) *Groups.* Compact connected groups acting transitively on $S^{d-1}$ with $d$ odd: $SO(d)$, and additionally $G_2$ for $d = 7$.

(iii) *Two bits.* In $\Omega_2 \otimes \Omega_2 \simeq \Omega_4$, (SA) applied twice gives a sub-bit $\Omega_2'$ containing $\omega_0 \otimes \omega_0$ and $\omega_1 \otimes \omega_1$. Local rotations $SO(d-1) \otimes SO(d-1)$ fixing the axis $\omega_0\omega_1$ preserve the $d$-dimensional span of $\Omega_2'$. For $d \geqslant 4$, the defining representation of $SO(d-1)$ is complex-irreducible, so $SO(d-1) \otimes SO(d-1)$ acts irreducibly on $\mathbb{R}^{d-1} \otimes \mathbb{R}^{d-1}$, which leaves no room for that subspace. Contradiction.

(iv) *The exception $d = 3$.* $SO(2)$ is Abelian and acts reducibly on $\mathbb{C}^2$, so the argument fails, consistently with quantum theory.

□

### 4.4 Higher capacities

*Proof sketch of Theorem 4.1.1.* (i) With the Bloch-ball identification $L$ for one bit, represent $\Omega_4$ via $L \otimes L$ inside $\mathrm{Herm}(\mathbb{C}^4)$. It contains all separable states.

(ii) Analysing the action of local phases on $\Omega_2'$ shows that its pure states are $|\psi\rangle\langle\psi|$ with $|\psi\rangle = \alpha|00\rangle + \beta|11\rangle$. Local unitaries then generate all pure two-qubit states.

(iii) By duality and the NRH, $\Omega_4$ is the full two-qubit state space, and the generated group is all of $U(4)$ acting by conjugation.

(iv) For $n$ bits, pairwise unitaries generate $U(2^n)$. Other capacities follow from (SA).

□

**Remark 4.4.1 (On the hypotheses):**

- Each hypothesis is used. Without (TL), real quantum theory survives (it is not locally tomographic). Without (CR), classical theory survives. Without (SA), other balls (e.g. $d = 7$ with $G_2$ for $n \geqslant 3$ bits) are not fully excluded.
- Alternative reconstructions: Hardy; Dakić–Brukner; Chiribella–D'Ariano–Perinotti (purification); and single-system postulates of spectrality and strong symmetry (Barnum–Hilgert).

## Chapter 5 Outlook: Links to the Other Notes

- *GPT and QRT.* A GPT supplies states, effects, channels and composites; a resource theory then selects free states and operations inside it. The categorical framework of the QRT notes, Chapter 0, takes the GPT's systems and channels as its process theory.
- *GPT and generalized contextuality.* Generalized (Spekkens) noncontextuality asks whether the operational statistics of a GPT, including its operational equivalences, admit a classical ontological model. For the allowed effects and states, this is equivalent to a *simplex embedding* of the GPT (Schmid et al.). The NRH and restricted theories matter here.
- *Higher-order interference.* Quantum theory has no third-order interference: $P_{123} = P_{12} + P_{13} + P_{23} - P_1 - P_2 - P_3$. GPTs with higher-order interference exist, and its absence can serve as an axiom (Barnum–Müller–Ududec).

## Appendix A Mathematical Preliminaries

**Definition A.0.1 (Cones):** A subset $C \subseteq V$ is a *convex cone* if $C + C \subseteq C$ and $\lambda C \subseteq C$ for all $\lambda \geqslant 0$. It is

- *pointed* if $C \cap -C = \{0\}$;
- *generating* if $C - C = V$.

A pointed cone defines a partial order $v \geqslant w \iff v - w \in C$, making $V$ an ordered vector space. The dual cone is $C^* = \{f \in V^* : f(C) \geqslant 0\}$.

**Theorem A.0.1: Separation and bipolar**

(i) If $C \subseteq V$ is a closed convex set and $x \notin C$, there is $f \in V^*$ with $f(x) < \inf_C f$.

(ii) For a closed convex cone, $C^{**} = C$.

(iii) $C$ is pointed iff $C^*$ is generating, and $C$ is generating iff $C^*$ is pointed (for closed $C$).

**Definition A.0.2 (Order unit, base):** $u \in C^*$ is a *strictly positive functional* if $u(C \setminus \{0\}) > 0$. Then $\{v \in C : u(v) = 1\}$ is a compact *base* of $C$, and $u$ is an *order unit* of $(V^*, C^*)$: for every $f \in V^*$ there is $\lambda$ with $f \leqslant \lambda u$.

**Definition A.0.3 (Tensor products of cones):**

$$C_A \otimes_{\min} C_B = \operatorname{cone}\{a \otimes b : a \in C_A,\ b \in C_B\},$$
$$C_A \otimes_{\max} C_B = (C_A^* \otimes_{\min} C_B^*)^*.$$

These are the cones over $\Omega_A \otimes_{\min} \Omega_B$ and $\Omega_A \otimes_{\max} \Omega_B$.

# Further Reading

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## D. Links to contextuality

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